

Sunidhi

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EDUCATION

Duke University (GPA 3.9/4.0)

May 2026

Master of Engineering in Biomedical Engineering- Medical Device Design

Durham, NC

Relevant Course Work: Advanced Manufacturing & Design, QMS, Biomaterials, Semiconductors, Embedded Systems Design, Design Health-NPD

EXPERIENCE

NPD Quality Engineer (Co-Op)

May 2025- January 2026

Johnson and Johnson MedTech

Irvine, CA

- Engineered an automated robotic actuator interfaced with the PFA generator, executing **100+** ablation cycles to eliminate manual testing and reduce validation cost by **\$38K**.
- Led attribute-based **DVT** testing across **270** trials (**30** catheters) using Minitab gageR&R and control charts, achieving **100%** accuracy and supporting final commercial release.
- Optimized polymer application by designing controlled experiments (**DOE**) to quantify material usage, and updated **SOPs** to reduce process variability by **17%** and enhance consistency.
- Detected catheter dimensional drift using Minitab normality testing, and Cp/Cpk analysis, driving targeted process changes that improved capability by **0.2** and reduced variation by **27%**.
- Investigated a shaft-break **NC** using Six Sigma **DMAIC**, identifying off-spec alloy composition as the source of failure and avoiding **~\$42K** in material scrap and rework costs.
- Designed a custom arcing test fixture to replicate high-voltage conditions and visualize discharge behaviour, enabling failure characterization.

Clinical R&D Engineer

October 2024 - April 2025

Duke Human Simulation and Safety Centre

Durham, NC

- Designed an ergonomic electrical cystoscope prototype, converting clinical design inputs into manufacturable architecture that reduced setup time by **20 minutes** per procedure.
- Applied **DFM** and tolerance optimization principles to improve injection-molded component quality and structural reliability, reducing manufacturing defects by **15%**.
- Conducted **DFMEA** and usability risk assessments across **15** failure modes, identifying design risks and implementing mitigations that reduced user-related defects and functional defects.

Sr Manufacturing Engineer

May 2022 - July 2024

S&M Industries

India

- Led ISO-controlled Class I–II cleanroom production across 2 lines and 4 product variants, delivering **~10,000** units/month while maintaining **>97%** FPY and **98%** OTIF through disciplined in-process quality controls.
- Established SPC-based process controls and capability monitoring, deploying automated fixtures that reduced cycle time by **18%** and increased equipment uptime by **12%**.
- Designed and executed **OQ** and **PQ** studies to establish critical process parameter ranges, using DOE and statistical analysis to reduce process variability by **19%**.
- Built and scaled a **14-person** cross-functional floor team, implementing training certification and layered process audits that improved audit readiness and reduced escalations by **30%**.
- Drove defect containment and disposition for line events through **CAPA** investigations and corrective actions, reducing defect rate by **35%** and scrap/rework by **27%** in one quarter.
- Implemented **SPC-based** process monitoring and capability tracking for device assembly, deploying fixtures and statistical controls that reduced assembly cycle time by **18%**.

PROJECT

Sleep-Apnea Surgical Device

September 2024 - April 2025

- Conducted **70+ hours** of clinical observation and patient interviews to identify key usability challenges in existing therapy, translating unmet clinical needs into design inputs for a tongue stabilization device.
- Designed and fabricated a functional prototype, conducted usability testing, demonstrating an improvement in patient comfort and consistent airway patency.
- Conducted **DFMEA** and **PFMEA** risk assessments across failure modes, identifying critical design and process risks and implementing mitigation strategies to improve device safety and manufacturability.

Spinal Biopsy and Aspiration Needle

November 2024 – December 2024

- Redesigned an **electromechanical** spinal biopsy and aspiration needle, incorporating a stable ergonomic handle with enhanced grip patterns, ensuring improved precision, control, and comfort during procedures.
- Integrated **optical spectroscopy sensors** into spinal biopsy and aspiration needles, enabling real-time tissue differentiation for enhanced precision and reduced procedural errors.

Medical Image Reconstruction for Mitral Valve

November 2024

- Processed and segmented high-resolution **DICOM** datasets using **3D Slicer** to reconstruct the mitral valve's 3D anatomy, enabling precise visualization and spatial orientation critical for device fitment.
- Designed a custom valve ensuring anatomical fit and conducted **FEA** analysis to confirm endurance under physiological conditions.

SKILLS

- Design and Stimulation** SolidWorks, AutoCAD, Fusion360, 3D-Slicer, Arduino, Design Control, 5whys, Fishbone analysis, GD&T, 7Ms
- Data Analysis** Six Sigma, SQL, Pandas, Minitab, Gage R&R, Verification & Validation, Risk Analysis, SPC, Tableau, SAP
- Quality & Regulatory** ISO 13485, FDA 510(k), Gage R&R, QMS (21 CFR Part 820), IEC 60601-1-6 / IEC 62366-1, MDR, IEC 62304, DHF